



Week 4 Status Report

Project: CCP-CN-S2026 – Problem 10: High Availability Router

Team Name: Guardian Gateways

Week2: (GNS3 Implementation & Basic Connectivity)

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Project Title: High Availability Router using HSRP

Tasks Completed:

- 1. Verified HSRP functionality between RouterA and RouterB.
- 2. Configured and tested SSH remote access.
- 3. Implemented ACL security policies.
- 4. Configured Port Security on switch ports.
- 5. Performed failover and recovery testing.
- 6. Verified network connectivity between all devices.
- 7. Collected performance statistics including latency and packet loss.
- 8. Saved final GNS3 project and device configurations.

IP Addressing Table

Device	IP
RouterA	192.168.10.2
RouterB	192.168.10.3
Virtual IP	192.168.10.1
PC1	192.168.10.10
PC2	192.168.10.20

Issues Faced:

- Initial SSH connectivity issue due to network addressing mismatch.
- ACL configuration required adjustment for management access.

Solutions Implemented:

- Corrected IP addressing and connectivity.

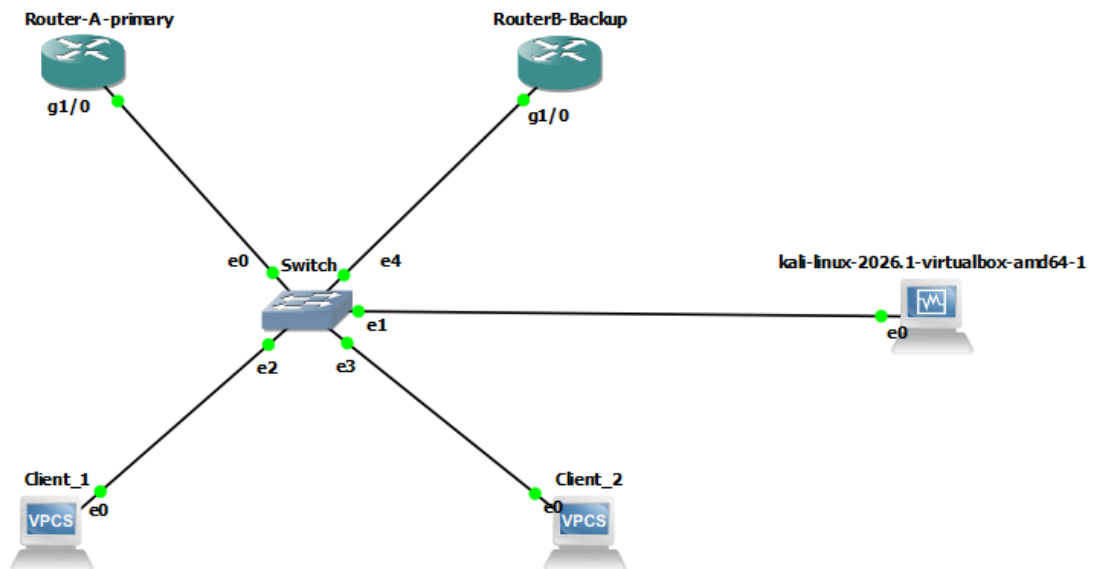
- Updated ACL rules to permit authorized management access.

Current Status:

All planned Week 4 tasks have been completed successfully. Network redundancy, security controls, and failover mechanisms are functioning as expected.

Overall Progress:

Project implementation completed and ready for final presentation and submission.



Verify Final HSRP Configuration:

```

RouterA#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State  Active        Standby        Virtual IP
Gi1/0      1    120 P Active local      192.168.10.3   192.168.10.1
RouterA#
  
```

```

Password:
RouterB#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State  Active        Standby        Virtual IP
Gi1/0      1    100 P Standby 192.168.10.2 local          192.168.10.1
RouterB#
  
```

Verify Interface Tracking:

```

G11/0      1      120 P Active local          192.168.10.3    192.168.10.1
RouterA#show track brief
Track  Object                Parameter        Value Last Change
1      interface GigabitEthernet1/0 line-protocol  Up    00:30:38
RouterA#

```

```

RouterB#show track brief
Track  Object                Parameter        Value Last Change
1      interface GigabitEthernet1/0 line-protocol  Up    00:30:16
RouterB#

```

Verify Routing Table:

```

RouterA#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet1/0
L       192.168.10.2/32 is directly connected, GigabitEthernet1/0

```

```

G11/0      1      100 P Standby 192.168.10.2    local          192.168.10.1
RouterB#show track brief
Track  Object                Parameter        Value Last Change
1      interface GigabitEthernet1/0 line-protocol  Up    00:30:16
RouterB#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
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Gateway of last resort is not set

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C       192.168.10.0/24 is directly connected, GigabitEthernet1/0
L       192.168.10.3/32 is directly connected, GigabitEthernet1/0
RouterB#

```

Save Final Configurations:

```
RouterB#copy running-config startup-config
Destination filename [startup-config]?
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
RouterB#
```

```
RouterA#copy running-config startup-config
Destination filename [startup-config]?
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
```

Verify ACL:

```
[OK]
RouterA#show access-lists
Standard IP access list 10
 10 permit 192.168.10.50
 20 deny any
Extended IP access list 101
 10 deny ip host 192.168.10.50 any
 20 permit ip any any log (816 matches)
RouterA#
```

```
RouterB#show access-list
Standard IP access list 10
 10 permit 192.168.10.50
 20 deny any
Extended IP access list 101
 10 deny ip host 192.168.10.50 any
 20 permit ip any any (8 matches)
```

```
RouterA#show access-list
Standard IP access list 10
  10 permit 192.168.10.50
  20 deny any
Extended IP access list 101
  10 deny ip host 192.168.10.50 any
  20 permit ip any any (13 matches)
RouterA#show ip interface g1/0
GigabitEthernet1/0 is up, line protocol is up
  Internet address is 192.168.10.2/24
  Broadcast address is 255.255.255.255
  Address determined by non-volatile memory
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Multicast reserved groups joined: 224.0.0.2
  Outgoing access list is not set
  Inbound access list is 101
  Proxy ARP is enabled
  Local Proxy ARP is disabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent
  IP fast switching is enabled
  IP Flow switching is disabled
  IP CEF switching is enabled
  IP CEF switching turbo vector
  IP CEF turbo switching turbo vector
  Associated unicast routing topologies:
--More--
```

Verify Login Security

```
RouterB#show login
  No login delay has been applied.
  No Quiet-Mode access list has been configured.

  Router NOT enabled to watch for login Attacks
```

Verify logging:

```

RouterA#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited, 0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

  Console logging: level debugging, 19 messages logged, xml disabled,
                    filtering disabled
  Monitor logging: level debugging, 0 messages logged, xml disabled,
                    filtering disabled
  Buffer logging:  level debugging, 19 messages logged, xml disabled,
                    filtering disabled
  Exception Logging: size (8192 bytes)
  Count and timestamp logging messages: disabled
  Persistent logging: disabled

No active filter modules.

  Trap logging: level informational, 22 message lines logged
--More--
*Jun 15 02:29:14.051: %SEC-6-IPACCESSLOGDP: list 101 permitted icmp 0.0.0.0 -> 192.168.10.2 (0/0), 9 packets

```

Ping Test:

```

VPCS> ping 192.168.10.20
84 bytes from 192.168.10.20 icmp_seq=1 ttl=64 time=0.871 ms
84 bytes from 192.168.10.20 icmp_seq=2 ttl=64 time=1.095 ms
84 bytes from 192.168.10.20 icmp_seq=3 ttl=64 time=0.733 ms
84 bytes from 192.168.10.20 icmp_seq=4 ttl=64 time=0.678 ms
84 bytes from 192.168.10.20 icmp_seq=5 ttl=64 time=0.737 ms

```

Continuous ping with very few packet losses in Kali.

```

(kali㉿kali)-[~]
$ ping 192.168.10.2
PING 192.168.10.2 (192.168.10.2) 56(84) bytes of data:
64 bytes from 192.168.10.2: icmp_seq=1 ttl=255 time=957 ms
64 bytes from 192.168.10.2: icmp_seq=2 ttl=255 time=173 ms
64 bytes from 192.168.10.2: icmp_seq=3 ttl=255 time=52.4 ms
64 bytes from 192.168.10.2: icmp_seq=4 ttl=255 time=10.1 ms
64 bytes from 192.168.10.2: icmp_seq=5 ttl=255 time=9.12 ms
^C
  192.168.10.2 ping statistics:
  5 packets transmitted, 5 received, 0% packet loss, time 4004ms
 rtt min/avg/max/mdev = 9.121/240.360/957.444/363.476 ms

```

Router1 Active and Router2 Standby

```

R1#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri  P State   Active      Standby      Virtual IP
Gi1/0      1    120 P Active  local       192.168.10.3 192.168.10.1
R1#

```

```
R2#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State   Active        Standby        Virtual IP
Gi1/0       1    100 P Standby  192.168.10.2   local          192.168.10.1
R2#
```

Client2 Ping to Client1(192.168.10.10)

```
VPCS> ping 192.168.10.10
84 bytes from 192.168.10.10 icmp_seq=1 ttl=64 time=0.838 ms
84 bytes from 192.168.10.10 icmp_seq=2 ttl=64 time=0.686 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=64 time=1.556 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=64 time=0.728 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=64 time=0.583 ms
```

Router B becomes Active.

```
Enter configuration commands, one per line. End with CNTL/Z.
RouterA(config)#interface g1/0
RouterA(config-if)#shutdown
RouterA(config-if)#
*Jun 15 03:07:26.487: %TRACKING-5-STATE: 1 interface Gi1/0 line-protocol Up->Down
*Jun 15 03:07:26.495: %HSRP-5-STATECHANGE: GigabitEthernet1/0 Grp 1 state Active -> Init
*Jun 15 03:07:28.483: %LINK-5-CHANGED: Interface GigabitEthernet1/0, changed state to administratively down
*Jun 15 03:07:29.483: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0, changed state to down
```

```
R2#
R2#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State   Active        Standby        Virtual IP
Gi1/0       1    100 P Active   local          unknown         192.168.10.1
R2#show s
*Jun 13 01:03:29.323: %HSRP-5-STATECHANGE: GigabitEthernet1/0 Grp 1 state Active -> Speaktand
GigabitEthernet1/0 - Group 1
```

Router A becomes Active.

```
R1#
R1#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State   Active        Standby        Virtual IP
Gi1/0       1    120 P Active   local          192.168.10.3   192.168.10.1
R1#
```

Continuous ping running:

```

VPCS> ping 192.168.10.20 -t
84 bytes from 192.168.10.20 icmp_seq=1 ttl=64 time=7.110 ms
84 bytes from 192.168.10.20 icmp_seq=2 ttl=64 time=1.199 ms
84 bytes from 192.168.10.20 icmp_seq=3 ttl=64 time=0.621 ms
84 bytes from 192.168.10.20 icmp_seq=4 ttl=64 time=0.673 ms
84 bytes from 192.168.10.20 icmp_seq=5 ttl=64 time=0.667 ms
84 bytes from 192.168.10.20 icmp_seq=6 ttl=64 time=0.922 ms
84 bytes from 192.168.10.20 icmp_seq=7 ttl=64 time=0.731 ms
84 bytes from 192.168.10.20 icmp_seq=8 ttl=64 time=0.886 ms
84 bytes from 192.168.10.20 icmp_seq=9 ttl=64 time=0.707 ms
84 bytes from 192.168.10.20 icmp_seq=10 ttl=64 time=0.950 ms
84 bytes from 192.168.10.20 icmp_seq=11 ttl=64 time=0.898 ms
84 bytes from 192.168.10.20 icmp_seq=12 ttl=64 time=0.680 ms
84 bytes from 192.168.10.20 icmp_seq=13 ttl=64 time=0.882 ms
84 bytes from 192.168.10.20 icmp_seq=14 ttl=64 time=0.669 ms
84 bytes from 192.168.10.20 icmp_seq=15 ttl=64 time=0.697 ms
84 bytes from 192.168.10.20 icmp_seq=16 ttl=64 time=0.683 ms
84 bytes from 192.168.10.20 icmp_seq=17 ttl=64 time=0.701 ms
84 bytes from 192.168.10.20 icmp_seq=18 ttl=64 time=0.674 ms

84 bytes from 192.168.10.20 icmp_seq=19 ttl=64 time=0.716 ms
84 bytes from 192.168.10.20 icmp_seq=20 ttl=64 time=0.657 ms
84 bytes from 192.168.10.20 icmp_seq=21 ttl=64 time=0.657 ms
84 bytes from 192.168.10.20 icmp_seq=22 ttl=64 time=0.686 ms
84 bytes from 192.168.10.20 icmp_seq=23 ttl=64 time=0.675 ms
84 bytes from 192.168.10.20 icmp_seq=24 ttl=64 time=0.780 ms
84 bytes from 192.168.10.20 icmp_seq=25 ttl=64 time=0.703 ms
84 bytes from 192.168.10.20 icmp_seq=26 ttl=64 time=0.694 ms
84 bytes from 192.168.10.20 icmp_seq=27 ttl=64 time=0.722 ms
84 bytes from 192.168.10.20 icmp_seq=28 ttl=64 time=0.754 ms
84 bytes from 192.168.10.20 icmp_seq=29 ttl=64 time=0.700 ms
84 bytes from 192.168.10.20 icmp_seq=30 ttl=64 time=0.646 ms
84 bytes from 192.168.10.20 icmp_seq=31 ttl=64 time=1.131 ms
84 bytes from 192.168.10.20 icmp_seq=32 ttl=64 time=0.673 ms
84 bytes from 192.168.10.20 icmp_seq=33 ttl=64 time=0.866 ms
84 bytes from 192.168.10.20 icmp_seq=34 ttl=64 time=0.685 ms
84 bytes from 192.168.10.20 icmp_seq=35 ttl=64 time=1.202 ms
84 bytes from 192.168.10.20 icmp_seq=36 ttl=64 time=0.690 ms
84 bytes from 192.168.10.20 icmp_seq=37 ttl=64 time=0.668 ms
84 bytes from 192.168.10.20 icmp_seq=38 ttl=64 time=0.661 ms
84 bytes from 192.168.10.20 icmp_seq=39 ttl=64 time=0.692 ms
84 bytes from 192.168.10.20 icmp_seq=40 ttl=64 time=0.739 ms
84 bytes from 192.168.10.20 icmp_seq=41 ttl=64 time=0.625 ms

```

Show All-to-All Ping Matrix

Source \ Destination	R1 (192.168.10.2)	R2 (192.168.10.3)	Client1 (192.168.10.10)	Client2 (192.168.10.20)
Client1	Success	Success	–	Success
Client2	Success	Success	Success	–
Router R1	–	Success	Success	Success
Router R2	Success	–	Success	Success

Final Test Matrix

Test	Expected	Result
PC1 → PC2 Ping	Success	Pass
SSH RouterA	Success	Pass
SSH RouterB	Success	Pass
Failover	RouterB Active	Pass
Recovery	RouterA Active	Pass
ACL	Block Unauthorized	Pass